Subject Description Form

Subject Code	ITC6980
Subject Title	Artificial Intelligence Techniques for Solving Management Problems in Fashion Industry
Credit Value	3
Level	6
Pre-requisite / Co-requisite / Exclusion	ITC502 Quality Assurance in Textiles & Clothing (Pre-requisite)
Objectives	1. To provide the fundamentals and application of various artificial intelligence techniques to assist decision makers in tackling key management, quality and technical problems in the fashion industry 2. To develop the ability of identifying and formulating optimization model for the identified problems in fashion enterprises 3. To develop the ability of applying appropriate artificial intelligence techniques to solve the management, quality and technical problems and technical problems in fashion enterprise
Intended Learning Outcomes	Upon completion of the subject, students will be able to: a. Identify potential management, quality and technical problems of fashion enterprises where artificial intelligence techniques can be applied b. Formulate optimization model for the identified problems of fashion enterprises c. Apply appropriate artificial intelligence techniques to solve the identified problems and optimize decision making
Subject Synopsis/ Indicative Syllabus	Topic 1: Understanding key management and quality decision points in the apparel industry Topic 2: Fundamentals of artificial intelligence techniques for apparel management applications Topic 3: Formulation of optimization models for management related problems, such as plant location, apparel production order planning, scheduling, cut order planning, marker planning, spreading and cutting scheduling, production systems using artificial intelligence techniques

Teaching/Learning Methodology

The learning approach will focus on application of artificial intelligence techniques and identification and formulation of appropriate optimization models for different management related problems in apparel industry.

This subject will be delivered based on student-centred learning method. Detailed and comprehensive literature reviews and case studies are designed for the students.

Students are required to submit detailed reports and present the findings of literature review, and proposal for identifying one management related problem for optimization model formulation with the suggestion of appropriated artificial intelligence technique(s). Their deliverables will be used for continuous assessment.

Assessment Methods in Alignment with Intended Learning Outcomes

Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed					
		a	b	c			
1. Continuous Assessment							
a) Individual Assignments	20%	✓	✓	✓			
b) Case Studies	20%	✓	✓	✓			
2. Final Examination	60%	✓	✓	✓			
Total	100%						

Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:

As all continuous assignments, case studies and final examination questions are designed with the consideration of both fundamental artificial intelligence mechanisms and techniques and real-life management problems, the three learning outcomes can be assessed.

Student Study Effort Expected

Class contact:	
▪ Briefing and in-depth Discussion	20 Hrs.
▪ Case studies	6 Hrs.
Other student study effort:	
▪ Student Self-Study	55 Hrs.
▪ Coursework	43 Hrs.
Total student study effort	124 Hrs.

Reading List and References

W.K. Wong, Z.X. Guo, S.Y.S. Leung, Optimizing Decision Making in the Apparel Supply Chain Using Artificial Intelligence: From Production to Retail, Woodhead Publishing, 1st edition